
Algorithm 1 Convergence Test of a Series $\sum a_n$

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1: function TESTCONVERGENCE( $\sum a_n$ )
2:   if  $\left(\left|\frac{a_n}{b_n}\right| \nrightarrow \infty\right)$  then                                     ▷ (1) Comparison Test
3:     return TESTCONVERGENCE( $\sum b_n$ )
4:   else if  $(a_n \nrightarrow 0)$  then                                           ▷ (2) Divergence Test
5:     return "Divergent"
6:   else if  $\left(\lim_{n \rightarrow \infty} \left|\frac{a_{n+1}}{a_n}\right| < 1\right) \vee \left(\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1\right)$  then       ▷ (3) Ratio/Root Test
7:     return "Absolutely Convergent"
8:   else if  $\left(|a_n| = \frac{1}{n^p}\right) \wedge (p > 1)$  then                               ▷ (4)  $p$ -Series Test
9:     return "Absolutely Convergent"
10:  else if  $\left(\int_1^\infty |a_x| dx \nrightarrow \infty\right)$  then                               ▷ (5) Integral Test
11:    return "Absolutely Convergent"
12:  else if  $(a_n = (-1)^n |a_n|)$  then                                       ▷ (6) Alternating Series Test
13:    return "Conditionally Convergent"
14:  else                                                                    ▷ (7) Failure of Tests
15:    return "Divergent"

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Some explanatory paragraph before the columns.

Item 1.	3. Third item in a numbered list.	22. Second item in a numbered list.
Item 2.	4. Fourth item in a numbered list.	23. Third item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	5. Fifth item in a numbered list.	24. Fourth item in a numbered list.
Item 1.	6. Sixth item in a numbered list.	25. Fifth item in a numbered list.
Item 2.	7. Seventh item in a numbered list.	26. Sixth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	8. Eighth item in a numbered list.	27. Seventh item in a numbered list.
Item 1.	9. Ninth item in a numbered list.	28. Eighth item in a numbered list.
Item 2.	10. Tenth item in a numbered list.	29. Ninth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	11. First item in a numbered list.	30. Tenth item in a numbered list.
Item 1.	12. Second item in a numbered list.	31. First item in a numbered list.
Item 2.	13. Third item in a numbered list.	32. Second item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	14. Fourth item in a numbered list.	33. Third item in a numbered list.
Item 1.	15. Fifth item in a numbered list.	34. Fourth item in a numbered list.
Item 2.	16. Sixth item in a numbered list.	35. Fifth item in a numbered list.
A longer paragraph to demonstrate wrapping across the columns. It will balance across the columns nicely.	17. Seventh item in a numbered list.	36. Sixth item in a numbered list.
1. First item in a numbered list.	18. Eighth item in a numbered list.	37. Seventh item in a numbered list.
2. Second item in a numbered list.	19. Ninth item in a numbered list.	38. Eighth item in a numbered list.
	20. Tenth item in a numbered list.	39. Ninth item in a numbered list.
	21. First item in a numbered list.	40. Tenth item in a numbered list.
